INCH-POUND
MIL-DTL-915/48D
22 August 2002
SUPERSEDING
MIL-C-915/48C
29 September 1989

DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, FOR OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 2SWF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-915 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First	-	Copper, conductor, in accordance with ASTM B 286, size 22-7, tin coated.
Second	-	Polyethylene insulation, natural, type II, class L, grade 4 in accordance with L-P-390. Nominal thickness 0.010 inch. Colored insulation, one black and one white conductor for each pair.
Third	-	Clear polyamide jacket on each conductor in accordance with ASTM D 4066. Nominal thickness 0.003 inch.
Fourth	-	Two conductors, one black and one white, cabled together with nominal lay of 2.5 inches to form each pair.
Fifth	-	Binder tape over each pair, at manufacturer's option, applied helically with overlap.
Sixth	-	Braided shield of AWG no. 34 or no. 36 tin coated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.

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Seventh - Shield insulation of one polyester tape, in accordance with NEMA FI 4 plus a jacket of clear polyamide in accordance with ASTM D 4066. Nominal thickness of 0.003 inch. Standard identification code applied by method 2.
- Alternate shield insulation of two polyester tapes, in accordance with NEMA FI 4, sealed. Standard identification code applied by method 2, on inner tape.

Eighth - The required number of pairs (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. l, from center outward. Nonfibrous fillers shall be employed.

Ninth - Synthetic rubber compound filled binder tape applied helically with overlap. (Surface of tape shall be such that cable jacket will adhere to it.)

Tenth - Arctic type polychloroprene jacket (black). (See table I for thickness.) Cable surface marking required.

TABLE I. Details.

Type and size	Number of pairs	Cable jacket thickness minimum (inch)	Overall diameter	
			Minimum (inches)	Maximum (inches)
2SWF-3	3	0.100	0.600	0.625
2SWF-4	4	0.090	0.600	0.625
2SWF-7	7	0.110	0.780	0.815
2SWF-24	24	0.125	1.190	1.250

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	17.71
Voltage withstand - volts root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Insulation resistance – megohms-1000 feet, minimum	
Conductor to conductor	3000
Conductor to shield	1500
Shield to shield	100
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage 1000 lbf/in^2, in 6 hours, in^3, (test each shipping length)	Zero
Capacitance	
Mutual capacitance - at l MHz, pF/ft, maximum	30
Capacitance unbalance - percent, maximum	8
Characteristic impedance - at l MHz, ohms	75 ± 5

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Group B:

Attenuation - at 3 MHz, db/100 feet, maximum 3
Cold working (minus 54 ± 2°C) ..... No damage
Drip - at 75 ± l°C ..... Zero
Physicals (unaged)
    Insulation (conductor)
        Tensile strength - lbf/in^2, minimum 1400
        Elongation - percent, minimum 300
    Jacket (cable)
        Tensile strength - lbf/in^2, minimum 1800
        Elongation - percent, minimum 300
        Set - inch, maximum 3/8

Group C:

Physicals (aged)
    Insulation (conductor)
        Air oven
            Elongation - percent of unaged, minimum 60
Permanence of printing (jacket) - cycles, minimum 250
Cable filler removability ..... No failure
Shield-conformance to material coverage and construction ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

    Cable aging and compatibility (95 ± 3°C) ..... No failure
    Cold working (minus 54 ± 2°C) ..... No failure

UNIT ORDERING LENGTH:

All sizes 1000 feet (nominal).

PART OR IDENTIFYING NUMBER (PIN): The PIN consists of the letter M, the general specification number along with the specification sheet, and a dash number representing type and size. Refer to MIL-DTL-915 for an example.

M	**915**	**/48-**	**X-X**
Prefix for Military Specification	Specification Number	Specification Sheet Number	Type and Size

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Review activities:
Army – AR, CR
Navy - CG, EC
DLA - CC

Preparing activity:
Navy - SH
(Project 6145-2317-009)

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